

# Machine Learning Reconstruction of Accretion Histories from Nova Luminosity Evolution

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## Abstract

Accretion-rate variability is one of the key physical processes governing the evolution of accreting white dwarfs and recurrent novae. While luminosity variations can be observed directly, the underlying accretion-rate history is generally inaccessible. In this project, we investigate whether machine learning models can reconstruct the accretion history of an accreting white dwarf using only luminosity evolution obtained from numerical simulations.

The dataset was generated from a large suite of nova simulations computed using the MESA stellar evolution code. The final dataset contains more than  $2 \times 10^6$  samples extracted from 107 independent simulations of a  $1.0 M_{\odot}$  accreting white dwarf.

We benchmarked multiple machine-learning approaches including Random Forests, XGBoost, LightGBM, CatBoost, Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and an Astroconformer-inspired Transformer model. The best performance was achieved using a Random Forest model augmented with physically motivated derivative features, reaching an  $R^2$  score of 0.877.

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# 1 Introduction

Recurrent novae are interacting binary systems consisting of a white dwarf accreting hydrogen-rich material from a companion star. As accreted material accumulates on the white dwarf surface, thermonuclear runaway eventually occurs, producing a nova eruption.

The recurrence time, ignition mass, peak luminosity, and overall evolution of these systems depend strongly on the accretion-rate history.

In observational astronomy, luminosity evolution can be measured directly through photometric observations. However, the underlying accretion rate is generally not observable. This motivates the following question:

Can the accretion-rate history of a nova system be reconstructed from its luminosity evolution using machine learning?

The goal of this project is to investigate this question using simulated nova light curves generated with the MESA stellar evolution code and a variety of machine-learning algorithms.

# 2 Physical Simulations

The simulations used in this project were generated as part of my Master’s thesis work using the Modules for Experiments in Stellar Astrophysics (MESA) code.

The simulation suite consists of:

- White dwarf mass:

$$M_{\text{WD}} = 1.0 M_{\odot}$$

- Hydrogen-rich accretion
- Multiple nova cycles
- Variable accretion histories
- 107 independent simulations

A representative simulation is shown below.

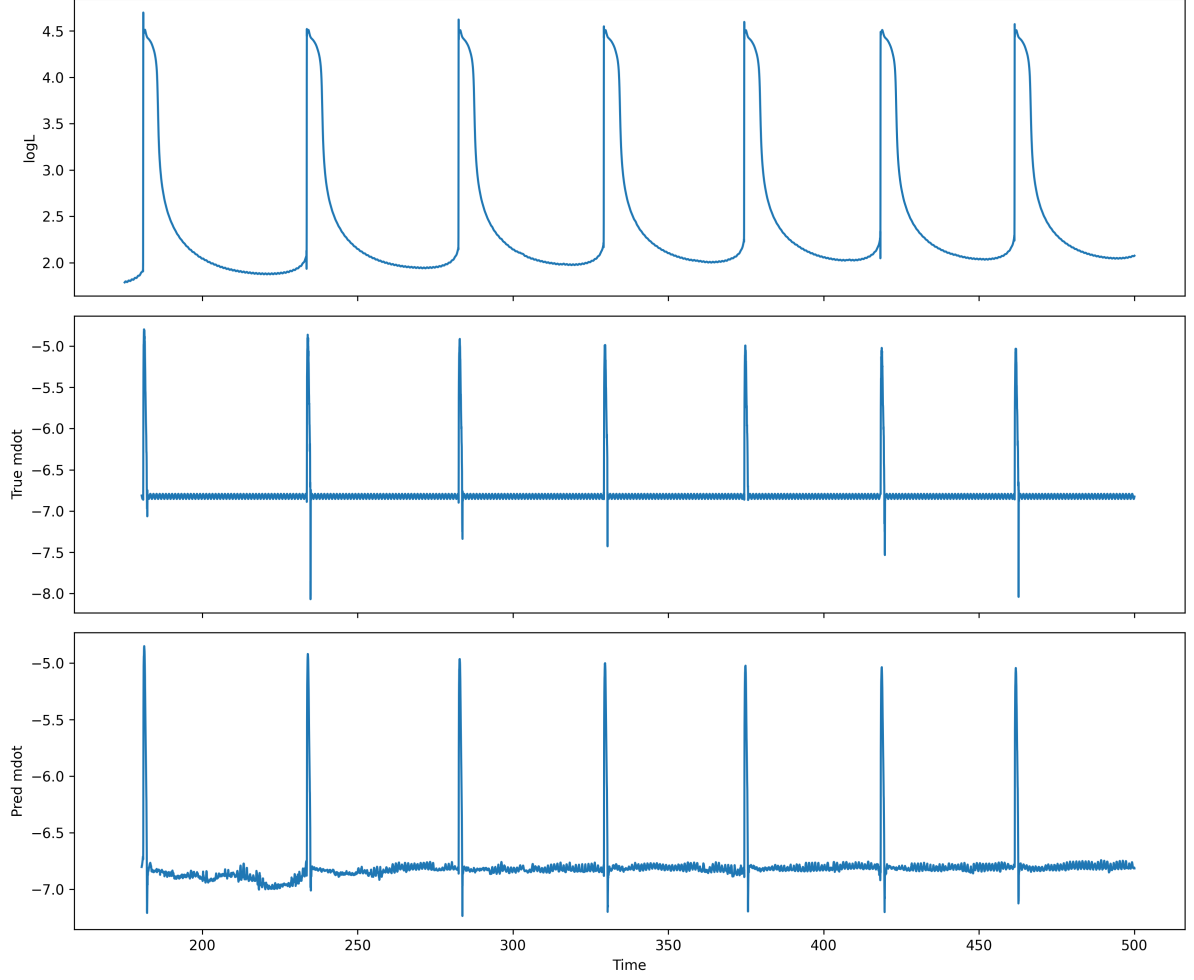


Figure 1: Evolution of stellar properties during a representative nova simulation generated using MESA.

The corresponding accretion-rate evolution is shown below.

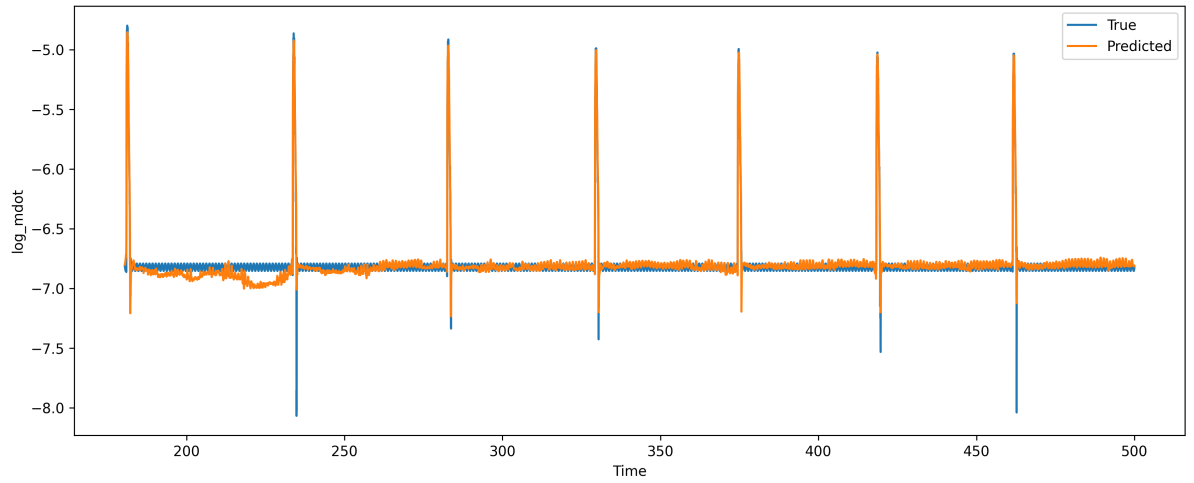


Figure 2: Accretion-rate history for the same simulation. The quantity  $\log \dot{M}$  serves as the machine-learning target.

### 3 Dataset Construction

The raw simulation outputs were interpolated onto uniform temporal grids and combined into a master dataset.

The final dataset contains approximately

$$N \approx 2.02 \times 10^6$$

samples obtained from

107

independent nova simulations.

Each row has the format

```
time, logL, log_mdot, sim_id, file_name
```

where:

- time = simulation time
- logL = logarithm of luminosity
- log\_mdot = logarithm of accretion rate
- sim\_id = simulation identifier
- file\_name = source file

A representative example is

```
time,logL,log_mdot,sim_id,file_name
175.422,4.216,-6.603,0,nova_clean_binned_data_78.csv
175.450,4.216,-6.603,0,nova_clean_binned_data_78.csv
175.477,4.216,-6.603,0,nova_clean_binned_data_78.csv
```

### 4 Train-Test Strategy

To prevent data leakage and ensure realistic performance estimates, the dataset was split at the simulation level rather than at the sample level.

Specifically,

80%

of simulations were used for training and

20%

were reserved for testing.

This ensures that the models are evaluated on entirely unseen nova simulations and therefore must generalize across different accretion histories.

Approximately:

- 85 simulations were used for training.
- 22 simulations were used for testing.

## 5 Window-Based Feature Construction

The machine-learning problem was formulated as a supervised regression task.

For each time step  $t$ , the target variable is

$$y = \log \dot{M}(t).$$

The input consists of a sliding luminosity window

$$X = [\log L_{t-W}, \dots, \log L_t].$$

Several window sizes were investigated:

$$W = 50, 100, 200, 400, 600, 800, 1000, 1200, 1400.$$

The primary benchmark models were trained using

$$W = 200.$$

## 6 Machine Learning Models

A broad range of machine-learning approaches was investigated.

### 6.1 Random Forest

Random Forest served as the baseline model.

The model achieved:

$$R^2 = 0.849.$$

### 6.2 Gradient Boosting Models

Three gradient-boosting algorithms were tested:

- XGBoost
- LightGBM
- CatBoost

These models produced performance comparable to Random Forest.

### 6.3 Artificial Neural Networks

Multiple ANN architectures were explored, including:

- 2 hidden layers
- 3 hidden layers
- 4 hidden layers

The best ANN achieved

$$R^2 = 0.814.$$

## 6.4 Convolutional Neural Networks

One-dimensional CNN architectures were trained using window sizes of:

$$50, 100, 200.$$

The best CNN achieved

$$R^2 = 0.829.$$

## 6.5 Astroconformer-Inspired Transformer

Motivated by recent Transformer architectures developed for astronomical time-series analysis, an Astroconformer-inspired model was implemented.

The model successfully learned the luminosity-accretion mapping but achieved

$$R^2 = 0.801.$$

which remained below the performance of the best tree-based methods.

# 7 Feature Engineering

A major focus of this project was the incorporation of physically motivated features.

## 7.1 First Derivative

In addition to luminosity history, we included

$$\frac{d \log L}{dt}$$

as an additional feature.

This doubles the feature count from 200 to 400.

The resulting Random Forest model achieved:

$$R^2 = 0.877,$$

$$\text{RMSE} = 0.0585,$$

$$\text{MAE} = 0.0269.$$

This represents the best result obtained in the project.

## 7.2 Second Derivative

We also investigated the inclusion of

$$\frac{d^2 \log L}{dt^2}.$$

Unexpectedly, performance deteriorated significantly:

$$R^2 = 0.755.$$

This suggests that higher-order derivatives introduce substantial numerical noise and provide little additional predictive information.

## 8 Results

| Model                       | RMSE   | MAE    | $R^2$ |
|-----------------------------|--------|--------|-------|
| RF + Derivative             | 0.0585 | 0.0269 | 0.877 |
| Random Forest               | 0.0647 | 0.0310 | 0.849 |
| XGBoost                     | 0.0653 | 0.0330 | 0.847 |
| LightGBM                    | 0.0653 | 0.0330 | 0.847 |
| CatBoost                    | 0.0675 | 0.0346 | 0.836 |
| CNN                         | 0.0690 | 0.0320 | 0.829 |
| ANN                         | 0.0719 | 0.0406 | 0.814 |
| Astroconformer              | 0.0744 | 0.0292 | 0.801 |
| RF + Derivative + Curvature | 0.1238 | 0.0474 | 0.755 |

Table 1: Summary of model performance.

## 9 Figures

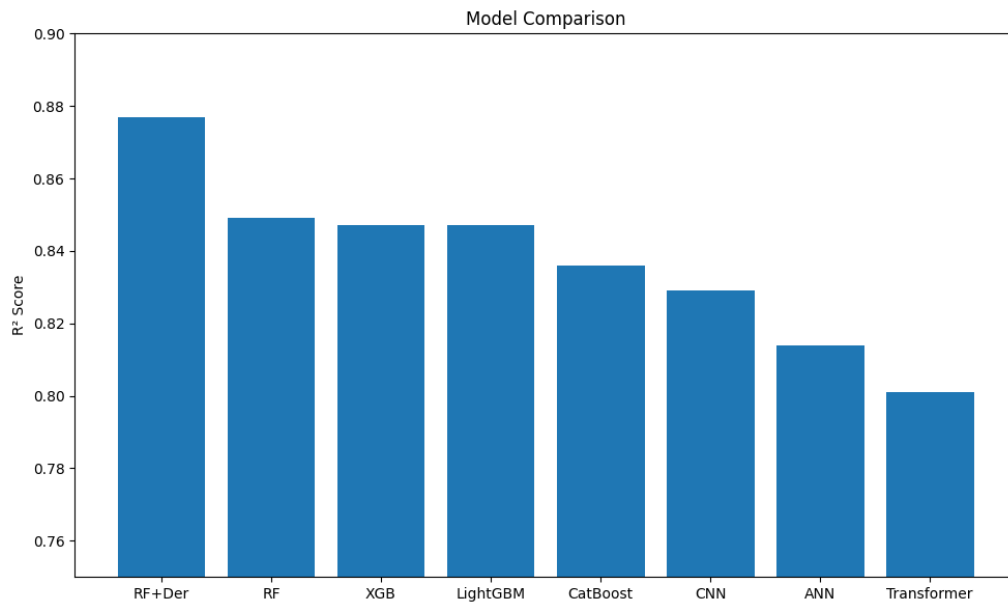


Figure 3: Comparison of all machine-learning models tested in this project.



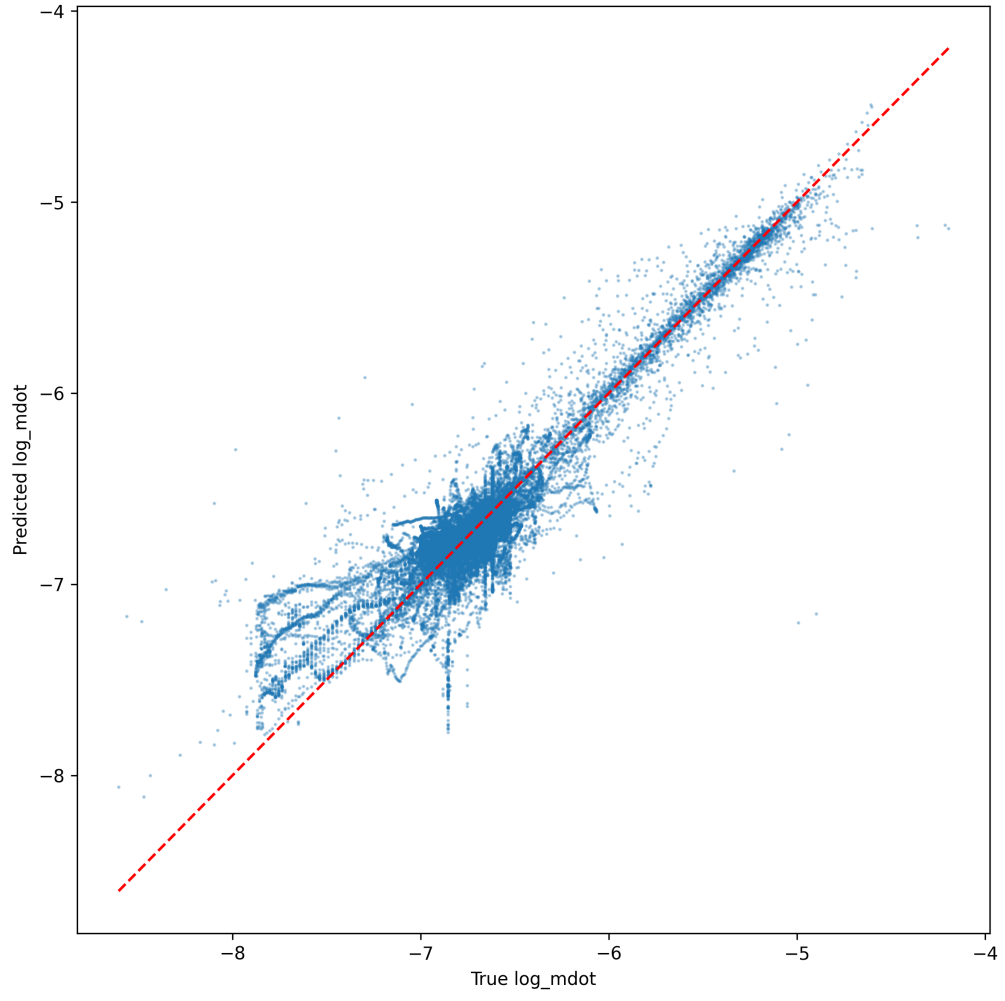


Figure 4: True versus predicted accretion rates for the best-performing RF + Derivative model.

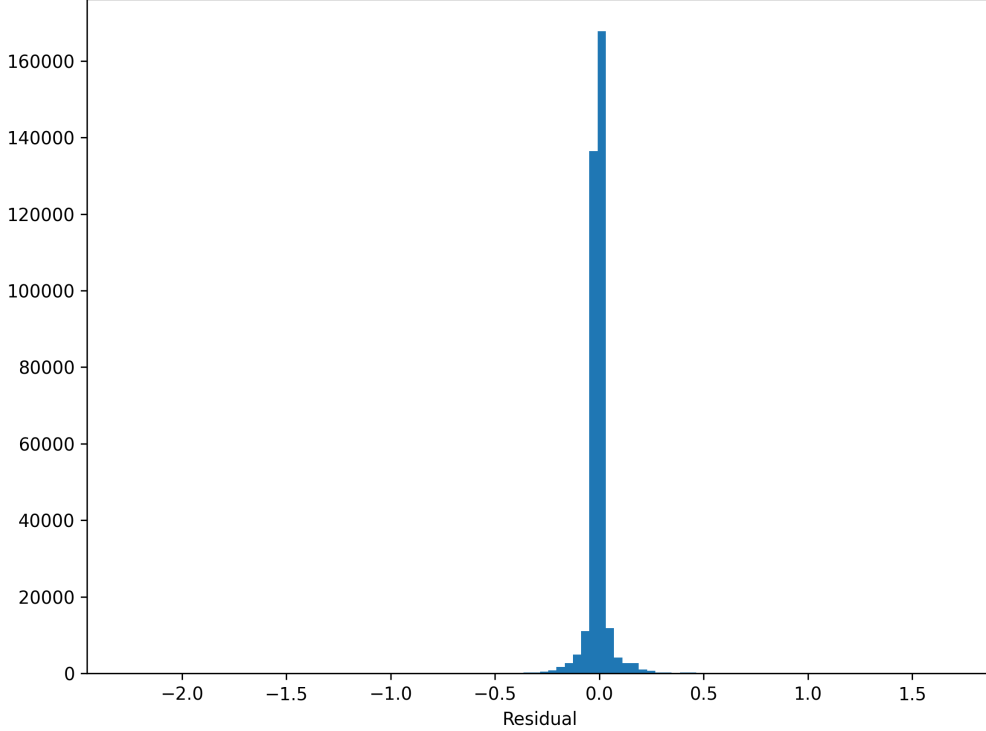


Figure 5: Residual distribution for the RF + Derivative model.

## 10 Discussion

Several important conclusions emerge from this study.

First, luminosity evolution contains substantial information regarding the underlying accretion history.

Second, feature engineering based on physical intuition produced larger performance gains than increasing model complexity.

Third, ensemble tree-based methods consistently outperformed deep-learning architectures for this particular dataset.

The strongest improvement came from incorporating the first derivative of luminosity, indicating that temporal evolution contains critical information about the accretion state.

## 11 Conclusion

This project demonstrates that machine-learning models can successfully reconstruct accretion-rate histories from nova luminosity evolution generated using MESA simulations.

The best-performing model, Random Forest combined with luminosity derivatives, achieved

$$R^2 = 0.877.$$

The results suggest that physically informed feature engineering is highly effective for this problem and provide a foundation for future studies involving broader white dwarf parameter spaces and observational datasets.

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